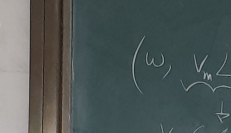
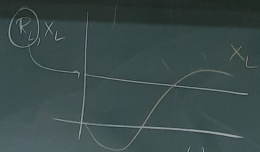
Resonance





Av power $\frac{1}{2} Re\{ v i^* \}$

Power from source: $\frac{1}{2} Re\{ V_s \, i^* \}$ → $i = \frac{V_s}{R_L + jX_L}$

Q: What is the power max ω?

ω dep

$\Rightarrow$ Q: What is the "ω" making $X_L = 0$?

$\Rightarrow$ "ω" when v & i are in phase?

$\frac{1}{2} Re\{ V_s i^* \} = \frac{|V_s|^2}{\sqrt{R_L^2 + X_L^2}} \cos\left(\tan^{-1} \frac{X_L}{R_L} \right)$

$\cos\theta = \cos(-\theta)$

$Re\left\{ \frac{|V_s|^2}{R_L - jX_L} \right\}$

$\left(\sqrt{R_L^2 + X_L^2} \angle -\tan^{-1} \frac{X_L}{R_L} \right)$

fixed

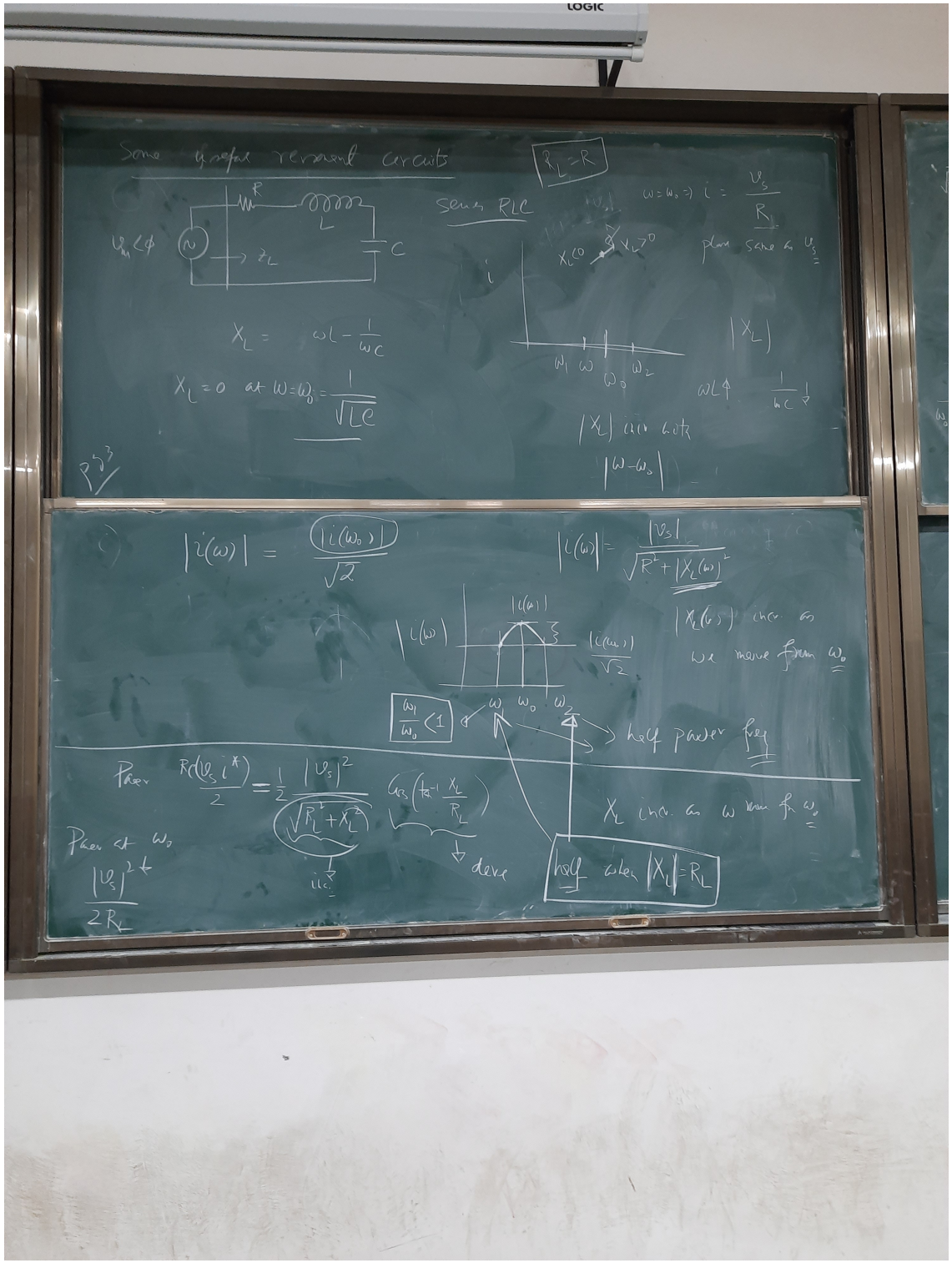
X_L depends with ω?

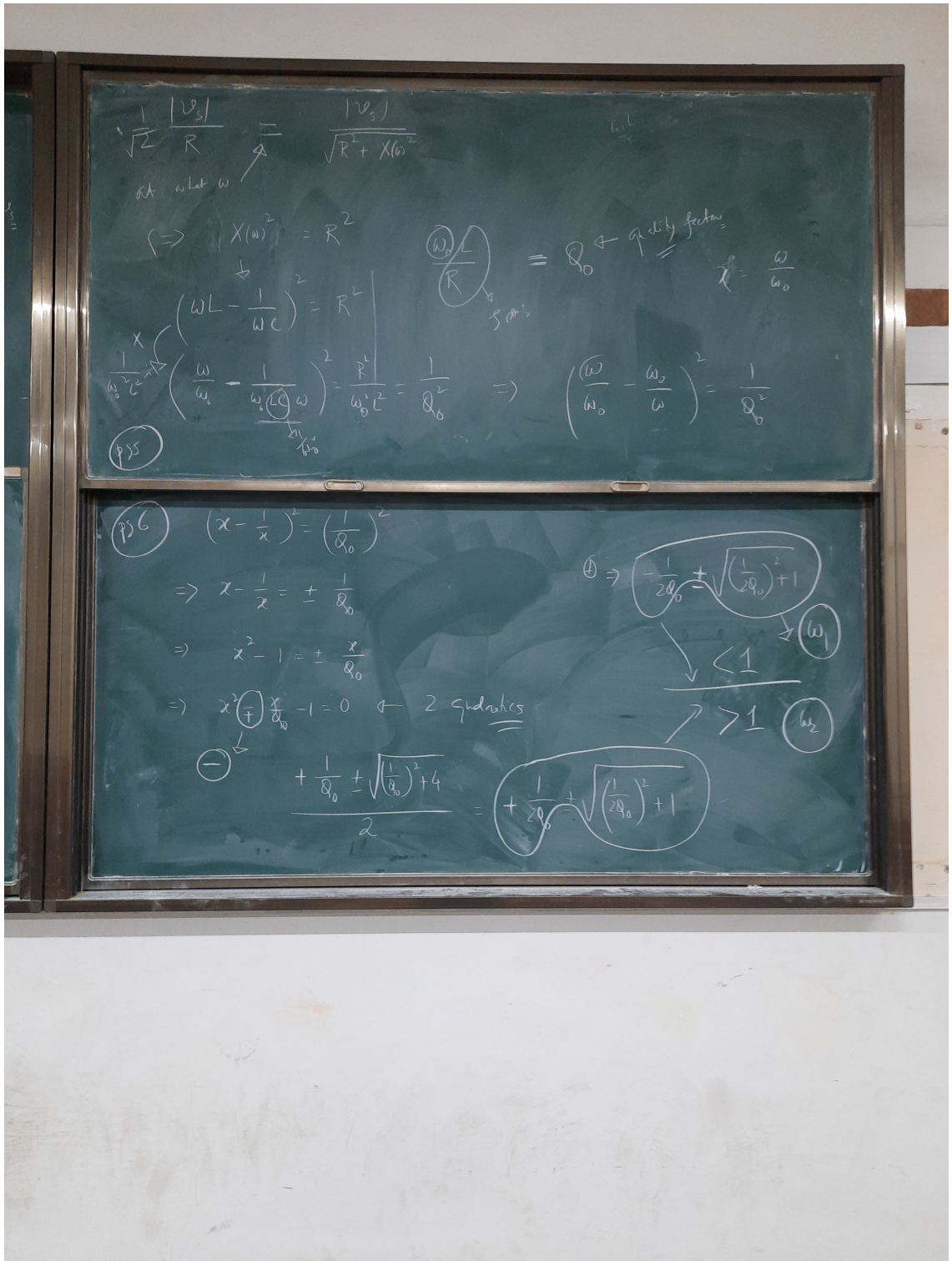
max if $X_L = 0$

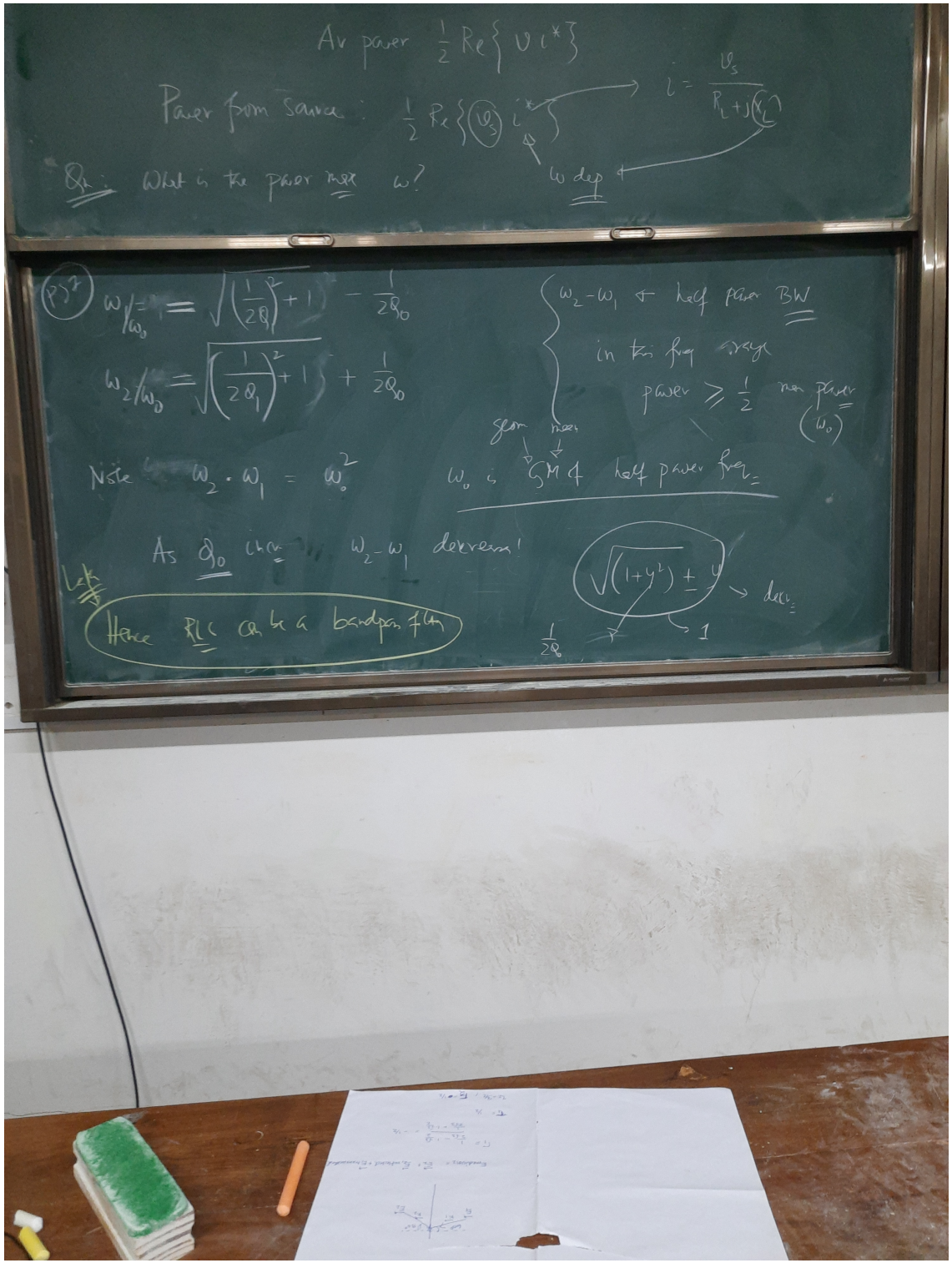
non-trivial

Note: For resonance we need both L & C !

O.w. $X_L = 0$ at a finite ω


Some useful resonant circuits
$R_L = R$
Series RLC
R
L
C
$V_s \angle \phi$
Z_L
$X_L = \omega L - \frac{1}{\omega C}$
$X_L = 0$ at $\omega = \omega_0 = \frac{1}{\sqrt{LC}}$
$\omega = \omega_0 \Rightarrow i = \frac{V_s}{R_L}$
$X_L < 0$ $V_L > 0$
plots same as V_s
$|X_L|$
ω_1 ω ω_0 ω_2
$\omega L \uparrow$ $\frac{1}{\omega C} \downarrow$
$|X_L|$ increases with $|\omega - \omega_0|$
$|i(\omega)| = \frac{|i(\omega_0)|}{\sqrt{2}}$
$|i(\omega)| = \frac{|V_s|}{\sqrt{R^2 + |X_L(\omega)|^2}}$
$|i(\omega)|$
$|i(\omega_0)|$
$\frac{|i(\omega_0)|}{\sqrt{2}}$
$|X_L(\omega)|$ increases as we move from ω_0
$\frac{\omega_1}{\omega_0} < 1$
half power freq
Power $\frac{R(V_s i^*)}{2} = \frac{1}{2} \frac{|V_s|^2}{\sqrt{R_L^2 + X_L^2}} \cos\left(\tan^{-1} \frac{X_L}{R_L}\right)$
decreases
X_L increases as ω moves from ω_0
Power at ω_0
$\frac{|V_s|^2}{2R_L}$
half when $|X_L| = R_L$


$\frac{1}{\sqrt{2}}\frac{|V_s|}{R} = \frac{|V_s|}{\sqrt{R^2 + X(\omega)^2}}$
at what ω
$\Rightarrow X(\omega)^2 = R^2$
$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$
$\frac{\omega_0 L}{R} = Q_0$ ← quality factor
$x = \frac{\omega}{\omega_0}$
$\left(\frac{\omega}{\omega_0} - \frac{1}{\omega_0 LC\,\omega}\right)^2 = \frac{R^2}{\omega_0^2 L^2} = \frac{1}{Q_0^2} \Rightarrow \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)^2 = \frac{1}{Q_0^2}$
p35
p36
$\left(x - \frac{1}{x}\right)^2 = \left(\frac{1}{Q_0}\right)^2$
$\Rightarrow x - \frac{1}{x} = \pm\frac{1}{Q_0}$
$\Rightarrow x^2 - 1 = \pm\frac{x}{Q_0}$
$\Rightarrow x^2 \mp \frac{x}{Q_0} - 1 = 0$ ← 2 quadratics
$+\frac{1}{Q_0} \pm \sqrt{\left(\frac{1}{Q_0}\right)^2 + 4}$
2
$= \left(+\frac{1}{2Q_0} \pm \sqrt{\left(\frac{1}{2Q_0}\right)^2 + 1}\right)$
$\Rightarrow \left(-\frac{1}{2Q_0} \pm \sqrt{\left(\frac{1}{2Q_0}\right)^2 + 1}\right)$
ω_1
< 1
> 1
ω_2


Av power $\frac{1}{2} Re\{ v i^* \}$
Power from source: $\frac{1}{2} Re\{ v_s i^* \}$
$i = \frac{v_s}{R_L + j(X_L)}$
Q: What is the power max w?
w dep
$\omega_1/\omega_0 = \sqrt{\left(\frac{1}{2Q_0}\right)^2 + 1} - \frac{1}{2Q_0}$
$\omega_2/\omega_0 = \sqrt{\left(\frac{1}{2Q_0}\right)^2 + 1} + \frac{1}{2Q_0}$
$\omega_2 - \omega_1$ ↔ half power BW
in this freq range
power ≥ $\frac{1}{2}$ max power (ω_0)
geom mean
Note $\omega_2 \cdot \omega_1 = \omega_0^2$
ω_0 is GM of half power freq
As Q_0 ↑ $\omega_2 - \omega_1$ decreases
$\sqrt{(1+y^2)} \pm y$ → decr
$\frac{1}{2Q_0}$
1
Hence RLC can be a bandpass filter